

Where Major Theorems Live in Proof Space

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A pilot map against LeanDojo Benchmark 4

Abstract

We place 18,852 declarations from nine major Lean formalization projects into statement and proof embedding spaces anchored by a fixed sample of 10,000 LeanDojo theorem–proof pairs. The frontier declarations are substantially more isolated than the reference corpus: 62.7% of their statement neighborhood densities fall below the fifth percentile of LeanDojo’s own density distribution. They are nevertheless not detached from the library. LeanDojo statements selected as semantic neighbors have proofs that are, on average, 0.0565 cosine points closer to the frontier proof than random references. The result supports a picture of frontier formalization as peripheral but locally connected to established proof infrastructure.

Corpus. The reference set $R = \{1, \dots, 10,000\}$ is the seed-0 LeanDojo sample used in the earlier semantic-geometry experiments. The frontier set contains declarations from OpenAI’s Ten Proofs and Navier–Stokes/Euler repositories, the complete FLT certificate, PFR, PNT+, Formalized Formal Logic, Jordan–Schönflies, Liquid Tensor, and Sphere Eversion. Large projects are represented by a seed-0 sample of approximately 2,500 declarations plus every identified headline theorem; smaller projects are retained in full. FLT is a stratified 2,003-declaration sample of theorem cards, proof modules, and definitions. This balancing prevents the largest generated repositories from determining the aggregate by sheer file count.

Frozen semantic map. For view $v \in \{s, p\}$, let $r_i^{(v)} \in \mathbb{R}^{1024}$ be the unit-normalized embedding of LeanDojo reference i , where s is theorem-statement text and p is proof text. Let $x_j^{(v)}$ denote the corresponding embedding of frontier declaration j . All vectors use the same frozen Cohere Embed v4 model and prompts as the original experiment; neither the reference sample nor its fitted clusters is changed.

For a frontier vector x define its exact top- k reference neighborhood and local reference density by

$$N_k^{(v)}(x) = \arg \operatorname{topk}_{i \in R} \langle x^{(v)}, r_i^{(v)} \rangle, \quad d_k^{(v)}(x) = \frac{1}{k} \sum_{i \in N_k^{(v)}(x)} \langle x^{(v)}, r_i^{(v)} \rangle.$$

Cosine similarity is the inner product because the vectors have unit length. For each reference vector, the same statistic is computed after excluding the vector itself. If $q_{.05}^{(v)}$ is the fifth percentile of those reference densities, the indicator

$$I^{(v)}(x) = \mathbf{1} \left[d_{10}^{(v)}(x) < q_{.05}^{(v)} \right]$$

marks a declaration as locally sparse relative to LeanDojo.

To test whether statement proximity carries proof information, let $S_j = N_{10}^{(s)}(x_j)$ and let U_j be ten seed-0 random reference indices. The transfer statistic is

$$\Delta_j = \frac{1}{10} \sum_{i \in S_j} \langle x_j^{(p)}, r_i^{(p)} \rangle - \frac{1}{10} \sum_{i \in U_j} \langle x_j^{(p)}, r_i^{(p)} \rangle.$$

This is a simple global-random baseline, not the context-matched baseline used in the earlier within-LeanDojo experiment.

Results. At $k = 10$, the frontier statement-density mean is 0.6717, compared with 0.7662 for reference declarations. The corresponding proof values are 0.7030 and 0.7971. Consequently, 62.7% of frontier statements and 60.7% of frontier proofs fall below their reference fifth-percentile thresholds. Isolation is therefore a corpus-level shift, not a property of only a few famous endpoints.

The project ordering is interpretable. The diverse 2026 Ten Proofs bundle and the new PDE corpus are the most peripheral, while logic and established topological developments lie closer to material already represented in Mathlib. Headline results need not be the most isolated parts of their projects: the FLT endpoint has nearest-statement

Project	n	$\bar{d}_{10}^{(s)}$	$I^{(s)}$ (%)	$\bar{\Delta}$
Ten Proofs	2,512	0.6437	80.0	0.0574
Navier–Stokes/Euler	2,507	0.6432	79.2	0.0513
PNT+	2,503	0.6435	73.0	0.0630
Jordan–Schönflies	2,501	0.6684	68.7	0.0499
PFR	910	0.6795	65.3	0.0657
FLT	2,003	0.6726	64.1	0.0590
Liquid Tensor	2,502	0.7044	42.1	0.0596
Sphere Eversion	909	0.7028	40.9	0.0590
Formalized Logic	2,505	0.7122	39.4	0.0520

Table 1: Project-level statement density, fraction below the LeanDojo fifth-percentile threshold, and statement-to-proof transfer advantage. Lower density and larger $I^{(s)}$ indicate a more peripheral project; positive $\bar{\Delta}$ means its statement neighbors retain useful proof-space signal.

cosine 0.7807, whereas a Navier–Stokes breakdown endpoint is as low as 0.5611. This distinction is why the project cloud is more informative than one dot per famous theorem.

Across all declarations, $\mathbb{E}[\Delta_j] = 0.0565$ (median 0.0524). Thus the frontier is sparse relative to LeanDojo but remains locally tethered to it: nearby established statements tend to have more relevant proof representations than arbitrary library statements. Direct overlap between the ten nearest statement and proof neighbors is low (mean Jaccard index 0.028), consistent with the earlier finding that statements and proofs share useful local structure without inducing the same global geometry. Figure 1 shows these two views together, allowing direct visual comparison of how the same projects occupy statement and proof space without changing the quantitative analysis.

Interpretation and limits. Major formalizations generally lie near the edge of the older LeanDojo corpus, yet remain built from locally recognizable semantic and procedural material. They are therefore useful out-of-distribution retrieval tests rather than wholly unrelated corpora. The statement result is primary: frontier proofs were extracted statically, while LeanDojo records tactic traces. Project comparisons also confound toolchain, domain, authorship, and AI use; embeddings measure textual semantics, the reference samples only 10,000 declarations, and the random baseline ignores source context. Confirmation requires native elaboration, dependency cones, and matched controls.

Reproducibility. The experiment freezes repository commits, sampling seed, preprocessing, embedding configuration, and a hard AWS spending cap. Inputs, embeddings, nearest-neighbor arrays, code, and machine-readable results are retained under `experiments/frontier-formalizations/`. Successful inference calls, including an abandoned larger pilot, cost an estimated \$0.60–\$0.96.

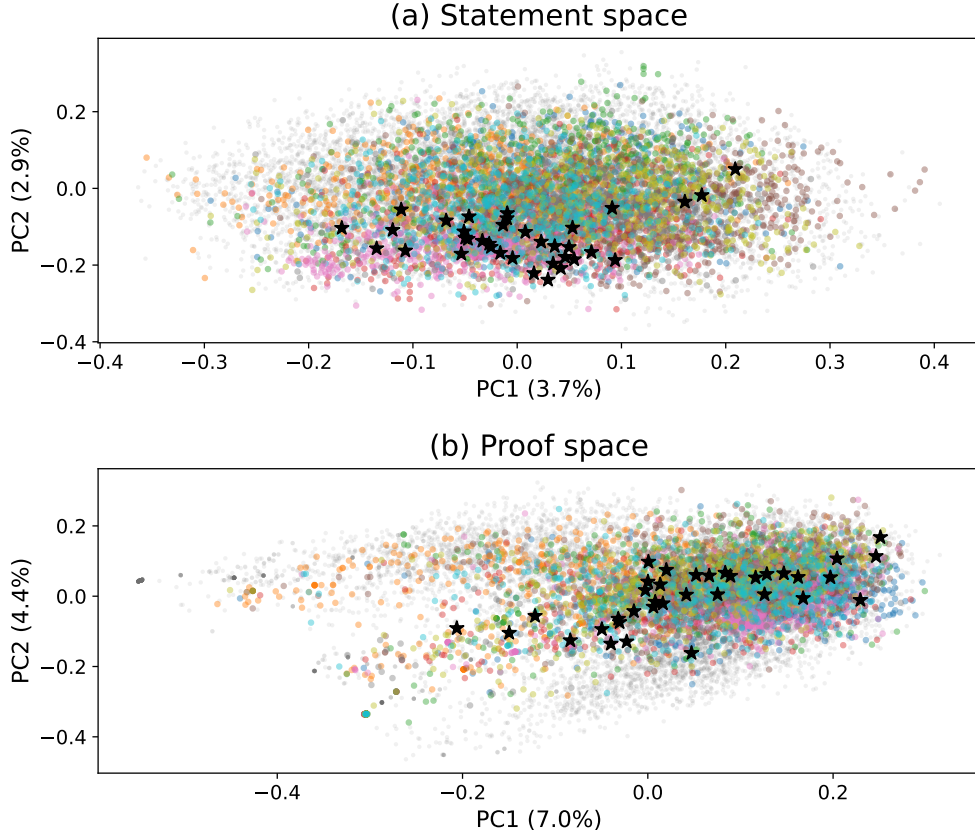


Figure 1: Statement (a) and proof (b) embeddings projected into separate 2D PCA bases fit only on the corresponding LeanDojo reference (gray). Color identifies the source project and stars mark headline declarations. Clouds displaced from the gray reference are visually peripheral; compact colored regions indicate project-specific local structure. The display is subsampled, and reported statistics use all 1,024-dimensional vectors.